MIL-S-1478F
31 December 1985
SUPERSEDING
MIL-S-1478E
29 December 1972

MILITARY SPECIFICATION

SNAP LINK, MOUNTAIN PITON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of steel snap link.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-W-470	- Wire, Steel, Carbon, Spring, Music
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8465

AMSC N/A

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads

DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

2-10-52	- Snap Link, Mountain, Piton

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 29	- Steel Bars, Carbon and Alloy, Hot Rolled and Cold-Finished, General Requirements For
A 322	- Hot Rolled Alloy, Steel Bars
A 331	- Steel Bars, Alloy, Cold-Finished
A 493	- Stainless and Heat-Resisting Steel for Cold Heading and Cold Forging-Bar and Wire
A 510	- Rods, Wire, and Coarse Round Wire, Carbon Steel, General Requirements For
B 633	- Electrodeposited Coatings of Zinc on Iron and Steel
D 3951	- Standard Practice for Commercial Packaging
E 18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials. (See 6.4).

* 3.2.1 Steel, alloy. Alloy steel for body and keeper (see 3.2.5) shall conform to steel number 4130, 4140, 41L30, 41L40, or a resulfurized grade of 41L40 (0.06 to 0.09 percent sulfur) of ASTM A 322 or ASTM A 331.

* 3.2.2 Steel, carbon. Carbon steel for pressure pin shall conform to grade number 1040 of ASTM A 510.

3.2.3 Wire, steel, spring. Spring steel wire for pressure spring shall conform to QQ-W-470.

* 3.2.4 Steel, stainless. Stainless steel for keeper pins shall conform to AISI type XM-7 (UNS designation S30430) of ASTM A 493.

3.2.5 Steel, alloy. Alternatively, steel for keeper shall conform to the chemical composition of table I and shall be tested as specified in 4.4.1.3.

TABLE I. Percentage of Chemical Composition

Alloy	Chemical Element				
AISI No.	C	Mn	P	S	Pb
12L14	.15 Max.	.85-1.15	.04-.09	.26-.35	.15-.35
11L17	.14-.20	1.00-1.30	.040 Max.	.08-.13	.15-.35

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* 3.3 Design and construction. The snap link shall conform to Drawing 2-10-52 and to the requirements specified herein. The keeper of the snap link shall travel from its fully closed position to its fully open position when a force of 2.0 pounds minimum to 5.5 pounds is applied in accordance with 4.5.1. When released, the keeper shall spring back to its fully closed position when tested as specified in 4.5.1, and there shall be no evidence of the keeper binding. The snap link shall withstand the tension test specified in 4.5.2 without breaking or cracking, and the keeper shall meet the performance requirements specified above after performance of the tension test. The keeper pins shall meet the peening requirements specified on Drawing 2-10-52 when tested as specified in 4.5.3.

* 3.4 Heat treatment. The body and keeper when fabricated from 4130, 4140, 41L30, 41L40, or resulfurized grade of 41L40 (0.06 to 0.09 percent sulfur) (see 3.2.1) shall be heat treated to a Rockwell hardness of C29 to C34 inclusive. The pressure pin shall be hardened to a Rockwell "15N" hardness of 78.5 to 83.0 inclusive. Heat treatment is not required for the keeper if made from steel alloys 12L14 or 11L17 (see 3.2.5).

3.5 Finish. All completed and unassembled component parts except keeper pins shall be zinc-coated in accordance with type I, classification FeZn 5 Sc 1, of ASTM B 633 prior to assembly.

3.6 Marking. The letters "US", the manufacturer's name or trademark, and the year of manufacture shall be legibly indented or etched prior to application of finish and located as shown on Drawing 2-10-52.

3.7 Workmanship. The finished snap link shall be free of burrs, sharp edges, flat areas, or corroded areas, and shall not be malformed. Etched or indented markings shall be applied so that no rough or sharp edges result.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required, the snap link shall be examined for defects specified in table II and dimensions specified in Drawing 2-10-52, and shall be tested as specified in 4.4.2.2. The presence of any visual defect, any dimensions not within the specified requirements, or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this document or applicable purchase document.

4.4.1.1 Dimensional examination of components. Any component dimension that is not within the tolerance specified on Drawing 2-10-52 shall be classified as a defect. A lot shall consist of all unplated components of a single type offered for examination at one time. The sample unit shall be one unplated component. The inspection level shall be S-3, and the acceptable quality level (AQL) shall be 2.5, expressed in terms of defects per hundred units.

4.4.1.2 Component hardness testing. Prior to assembly, the body, the keeper and the pressure pin (see 3.4) shall be tested for hardness in accordance with ASTM E 18. Rockwell C scale shall be used for the body and the keeper and Rockwell 15 N scale for the pressure pin. Superficial Rockwell hardness readings for the body and keeper shall be taken on 1/16-inch wide machined flat surfaces. These flat surfaces shall be located approximately 1 1/2 inches from either end and within the mid-length of body and near the mid-length of keeper at the thickest wall section. In lieu of using a component for hardness testing, test coupons may be processed with the heat treatment batch. The test coupons shall be of the same material as the components being heat treated. A lot shall consist of all bodies, keepers, or pressure pins, as applicable, heat treated in a single batch. The sample unit shall be one body, one keeper, or one pressure pin, or one test coupon, as applicable. Three determinations shall be made at each location on each

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sample unit and hardness shall be reported as the average of the three readings to the nearest appropriate Rockwell scale value. The inspection level shall be S-1, and the AQL shall be 2.5 for the body and for the keeper and 6.5 for the pressure pin, expressed in terms of defects per hundred units.

* 4.4.1.3 Alternative material, chemical analysis. Alternative steel for keeper referenced in 3.2.5 shall be tested and chemically analyzed in accordance with test requirements of ASTM A 29. Samples not in accordance with the chemical composition specified in 3.2.5 shall result in rejection of the lot of steel.

4.4.2 End item inspection. The inspection lot shall consist of all completely fabricated snap links offered for inspection at one time. The sample unit shall be one snap link.

* 4.4.2.1 Visual examination. The snap link shall be examined for defects in table II. The inspection level shall be II, and the AQL shall be 1.0 for major defects and 4.0 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of visual defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not smooth and adherent		201
	Missing	101	
Construction and workmanship	Not free of burrs, rough spots, sharp edges, slivers, flat areas, or projections	102	
	Any component missing or not specified type	103	
	Any component fractured, slit, punctured, dented, bowed, malformed or damaged	104	

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TABLE II. Classification of visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship - continued	Marking, etched or indented, contains rough or sharp edges	105	
	Any component loose	106	
	Keeper does not fully close due to binding caused by an interference fit or misalignment of keeper and body, or due to incorrect spring action	107	
	NOTE: To inspect for proper closing of keeper, open keeper and allow to return slowly (approximately two seconds). Any evidence of binding or incorrect spring action resulting in keeper not fully returning to closed positions shall be considered as a defect.		
	Keeper pin not peened correctly (i.e., peening not spread evenly or uniformly in the countersunk areas, peened end protrudes beyond surface of keeper, or metal breakout or cracks in peened area)	108	
Marking	Wrong location, incomplete, illegible, incorrect, or not accomplished as specified		202

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* 4.4.2.2 End item testing. When a first article is not required, the first produced snap link shall be subjected to all tests specified in table III. Failure of any test shall be cause for rejection of the first produced lot. Testing of the completely fabricated snap link shall be performed in accordance with table III. A one hundred percent inspection shall be made for the keeper opening test and closing test. The sample unit for the tests for tension and adequacy of keeper pins shall be one completely fabricated snap link

TABLE III. Testing of the end item

Characteristic	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported	Inspection level	AQL
Keeper opening and closing test	3.3	4.5.1	1	To nearest 0.1 pound	-	-
Longitudinal tension test	3.3	4.5.2	1	Pass or fail	S-2	2.5
Adequacy of peening of keeper pins test	3.3	4.5.3	4	Pass or fail	II	1.0

4.4.3 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with table IV. The lot size shall be the number of shipping containers offered for inspection at one time. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL shall be 2.5, expressed in terms of defects per hundred units.

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TABLE IV. Packaging inspection

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, or of improper size, location, sequence or method of application
Materials	Any component missing or damaged
Workmanship	Bulged or distorted container
Content	Number per container is more or less than required

* 4.4.4 Palletization inspection. An inspection shall be made to determine that the palletization complies with the secion 5 requirements. Defects shall be scored in accordance with table V. The lot size shall be the number of palletized unit loads in the inspection lot. The sample unit shall be one palletized unit load fully packaged. The inspection level shall be S-1, and the AQL shall be 6.5, expressed in terms of defects per hundred units.

TABLE V. Palletization inspection

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified

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TABLE V. Palletization inspection - Continued

Examine	Defect
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect, illegible, or of improper size, location, sequence, or method of application

4.5 Tests.

4.5.1 Keeper opening and closing test.

* 4.5.1.1 Apparatus. The apparatus (see figure 1) shall consist of a base plate on which is mounted a Chatillon force gage (or equal), a mounting plate for the snap link, and a guide to maintain proper alignment of the Chatillon force gage (or equal) rod. The mounting plate for the snap link shall rotate about a point directly below the keeper pin on which the keeper rotates when the snap link is on the plate, with the rod of the force gage applied to the keeper half way between the centers of the keeper pins with the major axes of the rod and keeper perpendicular. The snap link mounting plate shall be designed to hold the snap link in position during testing. It may be necessary to have an offset tip on the end of the rod to prevent the rod hitting the end of the snap link body before the keeper is completely open. Prior to use of the "or equal" gage, the contractor shall submit the gage to the contracting officer with supporting data for approval.

* 4.5.1.2 Procedure. Place the snap link on the rotating mounting plate with the rod and keeper positioned as described in 4.5.1.1, and rotate the mounting plate so the snap link keeper moves from the completely closed to the completely opened position and note the force required. Release the plate and allow the snap link to spring back to the completely closed position. Any nonconformance with the requirements of 3.3 shall constitute failure of the test and the item shall be rejected.

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* 4.5.2 Longitudinal tension test. The snap link shall be subjected to a 2,000 pound pull in the longitudinal direction, with the keeper in a closed position, on any industrial testing machine to determine compliance with 3.3. Any desired operating speed may be used until a load of 1,300 pounds is reached. After reaching that point, the speed shall be maintained at not more than 4 inches per minute nor less than 1 inch per minute until a load of 2,000 pounds is reached. This load shall be maintained at least 10 seconds, but no more than 15 seconds. Any cracking or breaking of link subjected to the tension pull shall constitute failure of this test. Snap links that do not break or crack shall be tested for keeper opening and closing as specified 4.5.1. Any nonconformance with the specified requirements shall constitute failure of the test.

* 4.5.3 Keeper pin adequacy of peening test. The snap link shall be placed in a vice or other suitable device and secured. Each keeper pin shall be tested for adequacy of peening by applying a force of 30 ± 3 pounds longitudinally on the end of the keeper pin. The test shall be performed on each end of both keeper pins. The force shall be applied with the probe of a force gage. The end of the probe shall be slightly smaller than the diameter of the keeper pin. Any longitudinal movement of either of the keeper pins shall constitute failure of the test.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Seventy snap links shall be unit packed in the most compact manner in a snug-fitting fiberboard box conforming to style RSC, type CF or SF, class domestic, grade 200 of PPP-B-636. Each box shall be closed in accordance with the appendix of PPP-B-636.

* 5.1.2 Commercial. Snap links shall be preserved in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Four hundred twenty snap links, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FTC, grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

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5.2.2 Level B packing. Four hundred twenty snap links, preserved as specified in 5.1, shall be packed in a snug-fitting, double-walled, fiberboard shipping container conforming to style FTC, type CF, class domestic, variety DW, grade 350 of PPP-B-636. Each shipping container shall be closed and reinforced as specified in 5.2.1.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636.

* 5.2.3 Commercial packing. Snap links, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), snap links, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The snap link is intended for use as a life-limb safeguard by mountain troops to secure a belaying rope to an anchored piton or similar application for helicopter lift missions and in fording streams.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
d. When weather-resistent fiberboard is required for level B shipments (see 5.2.2.1).
e. When palletization is required (see 5.3).

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* 6.3 First article. When a first article inspection is required, the item shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for inspection and approval of the first article.

* 6.4 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.2).

6.5 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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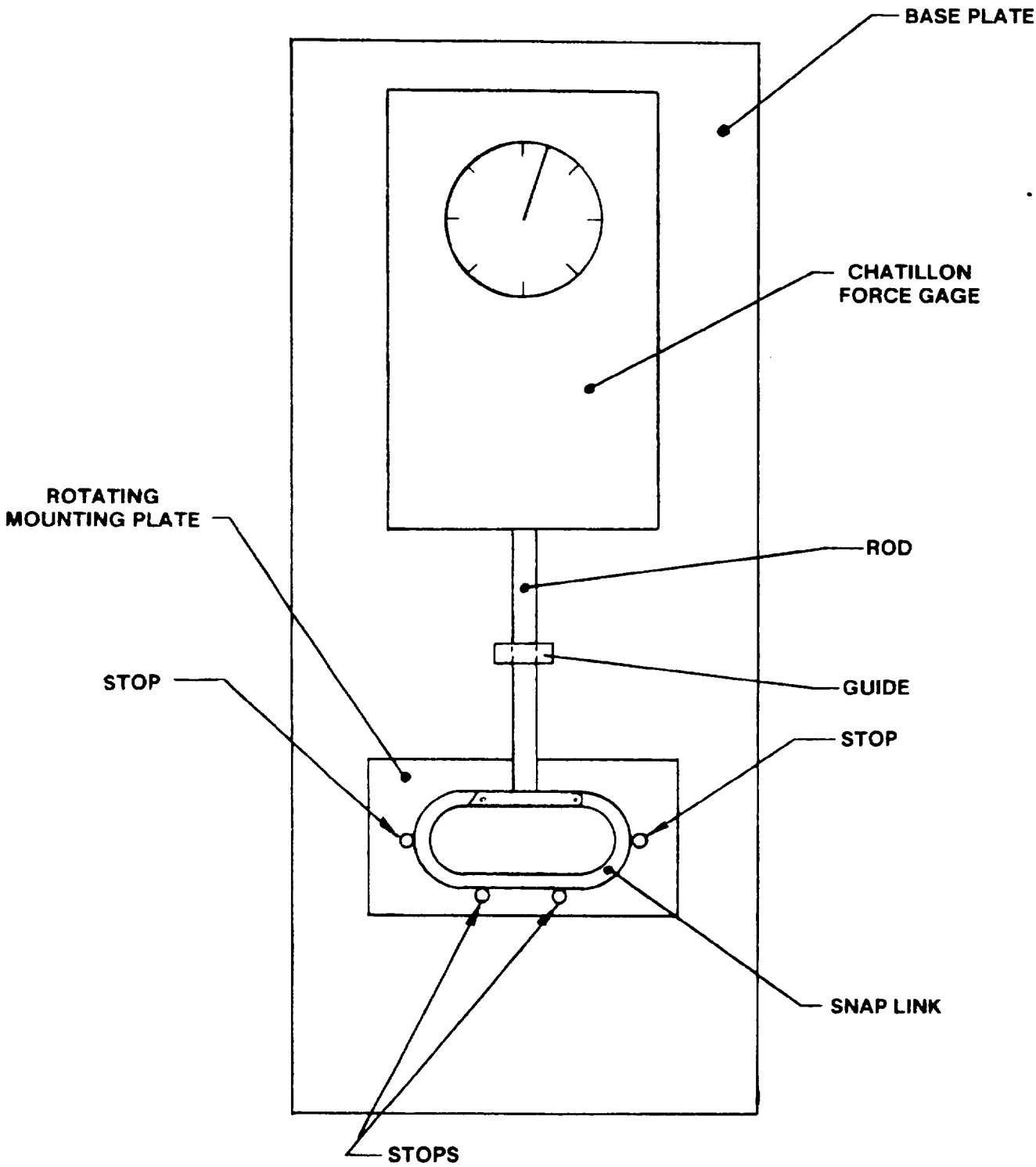


Figure 1. Snap Link Keeper Opening and Closing Jig

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-1478F	Snap Link, Mountain Piton

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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